ILP-Based Algorithm for Lithium-Ion Battery Charging Profile

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***Abstract*—A novel algorithm applying $(0,1)$-integer linear programming is proposed to search the optimal charging profile for rechargeable Lithium-ion batteries. The analysis of charging efficiency and charging time is based on equally segmented battery state of charge (SoC) and the results are re-assembled into a charging profile by $(0,1)$-integer linear programming. Experimental results show that the proposed algorithm achieves up to 18.25% and 21.38% charging time reduction in comparison to fast- and slow-charging specifications of a commercial product respectively, subject to a charging efficiency constraint at the same time.**

***Index Terms*—Lithium-ion battery charging profiles, multi-step constant current charging, $(0,1)$-integer linear programming.**

I. Introduction

Rechargeable Lithium-ion batteries become an essential power source for most of the prevalent portable electronic devices. The ability of the power source to maintain the operational functionalities as long as possible is very important in the user perception for purchasing portable electronic devices [1]. Rechargeable batteries with different electrochemical characteristics require different charging algorithms for better charging efficiency. To search an efficient scheme for recharging Lithium-ion batteries becomes an important topic for smart battery charger design [2] [3] [4]. Traditionally, a 3-phase constant current and constant voltage (CC-CV) process is applied for most of the commercial Lithium-ion chargers to charge a rechargeable battery, as shown in Fig.1. A pre-charged stage known as trickle charge is applied to the battery which is heavily discharged previously. During phase I, the battery is charged with a low current rate until the battery voltage reaches a threshold value to switching mode. It minimizes heat dissipation by avoiding to charge the battery with high current rate when the internal resistance of the battery is low at the initial stage. In phase II, a higher constant current rate is used to charge the battery till a pre-defined voltage level is reached, then a constant voltage mode is applied until the end of charge trigger condition is matched. Generally, the termination condition is set either when the charge current rate is below a pre-defined small value or the measured temperature is over a pre-set temperature. The CV mode is to avoid overpotential charging which can damage the battery cycle life [5].

The constant voltage (CV) mode dominates a major portion of battery charging time because the internal resistance builds up simultaneously when the battery voltage is rising. This can be shown in the exponential decreasing of the charge current during phase III. To decrease total recharging time, a multi-step constant current technique for phase II is proposed in [6] [7] for lead-acid and nickel/metal hybrid batteries. The studies showed convincing results in improving charging efficiency and reducing charging time for batteries used in electrical vehicles. Since multi-step constant current charging is feasible, the algorithm to search for an optimal charge current profile is also necessary. In [8], an ant colony system algorithm is applied to search the optimal charging profile for Lithium-ion batteries. Longer battery cycle life and faster charging time are shown in the studies.

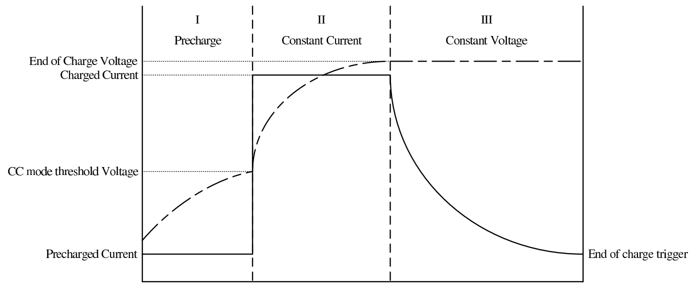

Fig. 1. Typical Charge Profile for Lithium-ion Batteries

In this study, a new searching algorithm using $(0,1)$-integer linear programming (ILP) is proposed to search for the optimal battery charging profile. A $(0,1)$-linear programming or binary linear programming is the special case of integer programming where variables are required to be 0 or 1 [9] [10]. $(0,1)$-ILP has been applied in many technical fields especially in branching problems. Here we model the search for optimal charging profile into a branching problem then apply $(0,1)$-ILP to search for the solution. The experimental results show up to 18.25% and 21.38% charging time reduction in comparison to fast-charging and slow-charging specifications of a commercial product while the charging efficiency is constrained to certain levels at the same time.

II. Battery Charging with Multi-Step Constant Currents

In a single-step constant current charging process, the temperature of the battery will rise up at the end of the charging

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stage if higher current rates are applied to the battery [11]. So, a medium charge current rate is often applied to the battery instead of a higher current rate. This is to prevent overpotential charging and overheating problems, but the charging time is then compromised. A multi-step constant current charging process can be regarded as a summation of several single-step constant current processes as shown in Fig. 2. The charge current rate at each stage is constrained to be lower than previous stages for a better charging efficiency, which is defined as the ratio of discharge capacity to the capacity charged into the battery during the charging process. At the same time, the temperature of the battery is confined in a safe range which is similar to the thermostatic charging process depicted in [11]. The capacity charged into the battery during the charging process can be formulated as the product of the charge current and the charging time. For the scheme of multi-step constant current and constant voltage, the total charged capacity Q_{total} is calculated as the sum of charged capacities for all constant current stages $\sum_j Q_j$ and the constant voltage stage Q_{cv}, as shown in Equation 1.

$$Q_{total} = \sum_j Q_j + Q_{cv} = \sum_j I_j t_j + \int I_{cv} dt \qquad (1)$$

For a battery charging process, it is a trade-off among the total charging time, the charging efficiency and the risk of over-potential and over-heating of the battery. If a high current rate (C-rate, in Ah or mAh) is adopted, the charging time is reduced but the charging efficiency is decreased and the risk of over-potential and over-heating is increased. On the other hand, if a lower current rate is chosen, the charging efficiency is increased but the total charging time is longer. Multi-step constant current and constant voltage scheme falls in between the two extreme case described above. In the beginning of the charging process, a higher current rate is adopted. As the capacity charged into the battery begins to build up, the lower current rates are applied step by step. Then the constant voltage is applied in the final stage until the end-of-charge condition is triggered. This scheme acquires a good balance between the total charging time and the charging efficiency, as well as reducing the risk of damage the cycle life of the battery. The studies in [6] [7] and [8] shows convinced results for this scheme and proposed algorithms for searching the current levels.

State-of-Charge (SoC) represents an estimated capacity residual inside a rechargable battery at the current instant of time. It is usually indicated in percentage $\%$ in comparison to the maximal amount of capacity that can be charged into the battery. The chemical dynamics inside the battery are complicated and many research activities have been conducted to obtain a good estimation of SoC [12]. For example, the measured battery voltage can be used to estimate rough SoC value for simplification. The charging efficiency of a battery, η, is defined as the ratio of discharged capacity to the capacity

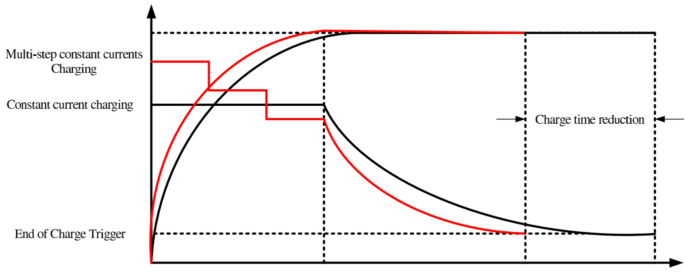

Fig. 2. Comparison of Single-Step Constant Current and Multi-Step Constant Current

previously charged into the battery, as shown in the Equation 2.

$$\eta = \frac{E_{discharge}}{E_{charge}} = \frac{\int V_{discharge} I_{discharge} dt}{\int V_{charge} I_{charge} dt} \qquad (2)$$

The battery charging efficiency at different stages of SoC is an important a priori information for the proposed searching algorithm. Battery charging and discharging tests at different C-rates and SoC stages are conducted to acquire the necessary data. In our experiment, the SoC is segmented evenly from 0% to 100% into equally segmented stages, e.g. $M1$, $M2$, $\ldots M10$, where $M1$ represents $0\% - 10\%$ SoC and $M10$ represents $90\% - 100\%$ SoC, etc. $1C$-rate is chosen to discharge the battery for all different charging C-rates to simulate the worst case discharging condition. The battery charging efficiency is computed based on this assumption. The proposed algorithm to search the optimal charging profile for Lithium-ion battery is to minimize the charging time subject to a constrained battery charging efficiency. This problem is then formulated as a branching problem and solved by $(0, 1)$-integer linear programming in Section III.

III. Integer Linear Programming for Optimal Charging Profile

Based on the analysis in Section II, the search for optimal charging profile is equivalent to finding the best charging current rate for each SoC segment. The search can then be modeled as a branching problem depicted in Fig. 3. The example in Fig. 3 demonstrates the space for all possible branches in the search. Here we apply $(0, 1)$-integer linear programming ($(0, 1)$-ILP or binary ILP) as a searching algorithm for optimal charging profile. $(0, 1)$-ILP is a special topic of linear programming which is applied broadly in branching problem [9] [10]. In this example, 10 charge current levels from $0.1C$ to $1C$ are assigned for multiple CC stages. For every charging current rate in a single SoC segment, the charge efficiency and charging time are acquired from charging and discharging tests. Then the optimal branch for minimal charging time subject to the constraint of charge efficiency is obtained from $(0, 1)$-ILP, which is described in details in the following sections.

A. Battery Charging and Discharging Tests

Battery charging and discharging tests are conducted by applying different constant charge rates for each SoC stage in order

TABLE I
CHARGING AND DISCHARGING TESTS FOR BATTERY EFFICIENCY

$\eta_(i,j)$	M=1	M=2	M=3	M=4	M=5	M=6	M=7	M=8	M=9	M=10
0.1C	90.52%	92.08%	92.09%	92.54%	92.75%	92.65%	92.73%	92.97%	93.12%	92.57%
0.2C	89.54%	89.35%	90.08%	90.67%	90.40%	90.94%	91.01%	91.03%	91.46%	91.89%
0.3C	87.98%	89.65%	89.68%	89.97%	90.20%	90.29%	90.49%	90.69%	90.77%	
0.4C	87.66%	89.16%	89.15%	89.44%	89.56%	89.67%	89.81%	89.96%	90.02%	
0.5C	87.15%	88.69%	88.64%	88.81%	88.93%	89.07%	89.15%	89.25%	89.46%	
0.6C	86.95%	88.40%	88.28%	88.17%	88.15%	88.47%	88.65%	88.75%	88.61%	
0.7C	86.13%	87.52%	87.50%	87.50%	87.62%	87.90%	88.10%	88.26%	88.22%	
0.8C	84.71%	87.04%	86.97%	87.04%	87.10%	87.22%	87.33%	87.46%	87.74%	
0.9C	84.88%	86.62%	86.61%	86.66%	86.73%	87.88%	87.27%	87.24%	87.51%	
1.0C	85.05%	86.20%	86.24%	86.28%	86.35%	86.53%	87.00%	87.02%	87.28%	

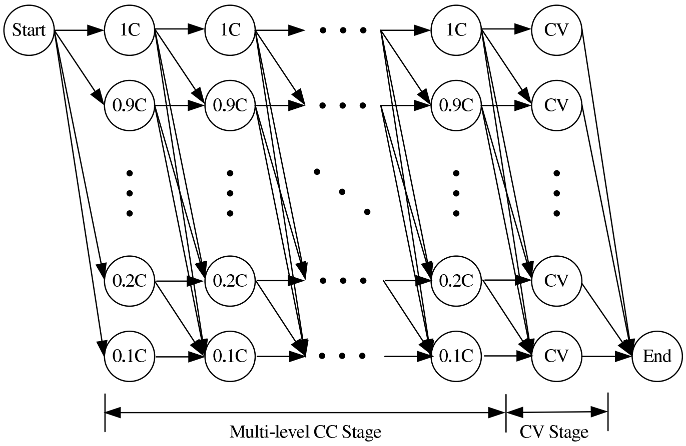

Fig. 3. Branch Searching Space of Optimization

to acquire a priori information about the charging efficiency of the battery. The main purpose of this test is to acquire battery efficiency for different charging currents. For example the charging currents range from $0.1C$ to $1C$ in this study. The data of battery charging efficiency for all kinds of charging current rates in the tests is calculated for each SoC segment and shown in TABLE. I.

B. ILP Formulation of multi-step constant current charging

The proposed charging scheme involves mutli-stage constant current charging and a constant-voltage charging in the last stage to avoid overpotential problems. To formulate the search for optimal charging profile into a branching problem and solve by $(0,1)$-ILP, we first introduce the nomenclature used in the modeling, as defined in TABLE. II.

Assume the feasible charging current level profile is as shown in Equation 3 which is a monotonically decreasing sequence. The reason to choose a monotonically decreasing sequence is that it preserves a better charging efficiency.

$$I_1 > I_2 > \cdots > I_N \tag{3}$$

The charging time of I_j in a single SoC segment is shown in Equation 4, and acquired by evenly segmenting the total capacity Q into M sections.

$$t_j = \frac{Q/M}{I_j}, j = 1 \dots N \tag{4}$$

As shown in Equation 5 and Equation 6, the discharging capacity is equal to the product of charging efficiency and the charging capacity, for both CC and CV modes. The values of charging efficiency required in the calculation are acquired from the charging and discharging tests in Section III-A. The discharging capacities in Equation 5 and Equation 6 will then be used in the calculation of total charging efficiency in Equation 14.

$$E_{dis}(i,j) = \eta_{i,j} E_{ch}(i,j), i = 1 \dots M, j = 1 \dots N \tag{5}$$

$$E_{cv,dis}(i) = \eta_{cv}(i) E_{cv,ch}(i), i = 1 \dots M \tag{6}$$

T_j represents the total period of time charged with the same current rate I_j.

$$T_j = S_j t_j, j = 1 \dots N \tag{7}$$

TABLE II
NOMENCLATURE IN OPTIMAL CHARGING PROFILE WITH ILP

Symbol	Description
C	battery current rate in Ah or mAh
Q	total battery capacity
M	the number of SoC segments
N	the number of charge current levels
I_j	the j_{th} charge current
t_j	the charging time for charging current I_j at a single SoC stage
η_j	the charging efficiency of charging from 0% to the relative SoC charging with I_j.
$\eta_{i,j}$	the charging efficiency when charging SoC_i with I_j
η_{cv}	the charging efficiency for constant-voltage state
$E_{dis}(i,j)$	the discharge capacity of CC stage with discharge current I_j
$E_{ch}(i,j)$	the charge capacity of CC stages with discharge current I_j
T_{CV}	the charging time of CV stages.
η_{CV}	the charge efficiency of CV stages.
$E_{cv,dis}(i)$	the discharge capacity of i_{th} CV segments.
$E_{cv,ch}(i)$	the charge capacity of i_{th} CV segments.
T_j	the charging time for charging current I_j
S_j	the number of SoC segments charged with I_j
$SSoC_j$	the starting SoC of I_j
T	the overall charging time

TABLE III
OPTIMAL CHARGING PROFILE WITH SINGLE STAGE CV

M1	M2	M3	M4	M5	M6	M7	M8	M9
1C	0.9C	0.8C	0.8C	0.8C	0.7C	0.7C	0.7C	0.3C

TABLE IV
BATTERY CHARGING SPECIFICATIONS

Capacity	Charging Mode	efficiency	Time
1000 mAh	CC-CV	88.04%	179.67 mins

$SSoC_j$ means the first SoC segment applied with current rate I_j and it is shown in Equation 8.

$$SSoC_j = 1 + \sum_{k=1}^{j-1} S_k, j = 2 \dots N \tag{8}$$

subject to

$$S_j \geq 0 \tag{9}$$

$$S_j \leq M - 1, j = 1 \dots N \tag{10}$$

$$\sum_{j=1}^{N} S_j \leq M - 1 \tag{11}$$

where $SSoC_1 = 1$. $SSoC_{cv}$ represents the last SoC segment for the CV stage. Equation 9 is a trivial constraint for the optimization. Equation 10 and Equation 11 confine that at least one CV stage is required to complete the charging process in order to prevent overpotential charging.

Equation 12 and Equation 13 represent the overall discharging and charging capacities for CC mode, respectively.

$$E_{cc,dis}(SSoC_{cc}) = \sum_{j \in S_j \neq 0} \sum_{j=SSoC_j}^{SSoC_j+(S_j-1)} E_{dis}(i,j) \tag{12}$$

$$E_{cc,ch}(SSoC_{cc}) = \sum_{j \in S_j \neq 0} \sum_{j=SSoC_j}^{SSoC_j+(S_j-1)} E_{ch}(i,j) \tag{13}$$

The overall charging efficiency in Equation 14 is obtained by the ratio of total discharging capacity to the total charging capacity for both CC and CV modes. η_{min} in Equation 15 is a pre-defined constraint for the charging efficiency η during the optimization process.

$$\eta = \frac{E_{cv,dis}(SSoC_{cv}) + E_{cc,dis}(SSoC_{cc})}{E_{cv,ch}(SSoC_{cv}) + E_{cc,ch}(SSoC_{cc})} \tag{14}$$

$$\eta \geq \eta_{min} \tag{15}$$

The purpose of the optimization is to minimize the total charging time T, where

$$T = T_{cv}(SSoC_{cv}) + \sum_{j=1}^{N} T_j \tag{16}$$

Suppose a set of charging current level is chosen as $\{1C, 0.9C, 0.8C, \dots, 0.1C\}$ and SoC segments are $\{0\% - 10\%, 10\% - 20\%, \dots, 90\% - 100\%\}$ (marked as $M1$, $M2$,...,$M10$), where 90% to 100% is applied with CV mode. The experimental results for both CC-CV and proposed multi-level CC and CV regimes are shown in TABLE. III, where $M10$ is the CV mode. The total charging time is 150 minutes and the overall charging efficiency is 88.051%. It shows a better result in the total charging time than the outcome of the battery applying CC-CV method in TABLE IV.

C. Modification for Multi-Step CC and CV Charging

In Section III, the last segment of charging is defined as constant voltage (CV) phase. This raises another question: is one single CV segment better than multiple CV segments? Here we analyze the feasibility of multiple CV segments. For this purpose, we modify the single CV stage into multiple-CV stages of the proposed searching space shown in Fig. 3 and run the $(0, 1)$-ILP to search the optimal profile. The $SSoC_{CV}$ is modified as shown in Equation 17.

$$SSoC_{CV} = 1 + \sum_{k=1}^{N} S_k \tag{17}$$

The standard ILP formulation for a Lithium-ion battery charging profile is shown in TABLE. V and the algorithm flow chart is described in Fig. 4. First, the number of charging current rates and the number of SoC segments are chosen. Then a branching space for $(0, 1)$-ILP is formed. Apply all feasible current sequences in the calculation of charging efficiency and charging time will end up with an optimal solution for the charging profile. The charging efficiency and the charging time for different numbers of CV stages are also calculated in TABLE. VI. The charging profile is shown in Fig. 5, and the charging efficiency and the total charging time are 88.046% and 146.87 minutes, respectively, as shown in TABLE. VII. The charging time is slightly shorter than the charging time shown in Section III-B.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

In this section, we apply the optimal charging profile acquired from the search in Section III by ILP method to a commercial Lithium-ion battery, and compare the result with its specification using CC-CV scheme. The platform for battery charging experiments conducted in this research is assembled by a IBM compatible PC, NI Labview software, TI gas gauge development suite and commercial off-the-shelf Lithium-ion batteries.

A. Analysis On The Influence of Number of SoC Segments and Current Levels

In this section, we analyze the effect of increasing numbers of SoC segments and current levels on the optimal charging time and charging efficiency. Four, five, ten, and twenty SoC

TABLE VII
OPTIMAL CHARGING STRATEGY AFTER MODIFICATION

M_1	M_2	M_3	M_4	M_5	M_6	M_7	M_8	M_9	M_{10}	η	Time(min)
1C	1C	0.9C	0.8C	0.7C	0.7C	0.7C	0.6C	CV	CV	88.0465%	146.87

TABLE V
$(0,1)$-ILP ALGORITHM FOR BATTERY CHARGING PROFILE

Given C, M, N, table of $eff(i,j)$ and $E_{ch}(i,j)$, table of $T_{cv}(i)$, $E_{cv,ch}(i)$ and $E_{cv,dis}(i)$ and the efficiency constraint η_{min},

$$I_1 > I_2 > I_3 > \cdots > I_N$$

$$t_j = \frac{Q/M}{I_j}, j = 1 \ldots N$$

$$E_{dis}(i,j) = \eta_{eff} E_{ch}(i,j), i = 1 \ldots M, j = 1 \ldots N$$

$$E_{cv,dis}(i) = \eta_{cv}(i) E_{cv,ch}(i), i = 1 \ldots M$$

$$T_j = S_j t_j, j = 1 \ldots N$$

$$SSoC_j = 1 + \sum_{k=1}^{j-1} S_k, j = 2 \ldots N$$

$$SSoC_{CV} = 1 + \sum_{k=1}^{N} S_k$$

subject to

$$S_j \geq 0$$

$$S_j \leq M - 1, j = 1 \ldots N$$

$$\sum_{j=1}^{N} S_j \leq M - 1$$

$$E_{cc,dis}(SSoC_{cc}) = \sum_{j \in S_j \neq 0} \sum_{j=SSoC_j}^{SSoC_j+(S_j-1)} E_{dis}(i,j)$$

$$E_{cc,ch}(SSoC_{cc}) = \sum_{j \in S_j \neq 0} \sum_{j=SSoC_j}^{SSoC_j+(S_j-1)} E_{ch}(i,j)$$

$$\eta = \frac{E_{cv,dis}(SSoC_{cv}) + E_{cc,dis}(SSoC_{cc})}{E_{cv,ch}(SSoC_{cv}) + E_{cc,ch}(SSoC_{cc})}$$

$$\eta \geq \eta_{min}$$

$$T = T_{cv}(SSoC_{cv}) + \sum_{j=1}^{N} T_j$$

Objective:

To minimize T.

TABLE VI
BATTERY EFFICIENCY FOR MULTI-SEGMENT CV MODE

CV Segments	M=6-10	M=7-10	M=8-10	M=9-10	M=10
η_{eff}	89%	90.1%	90.7%	91.6%	92.7%
$T(SSoC_{CV})$	150min	130min	110min	85min	70min

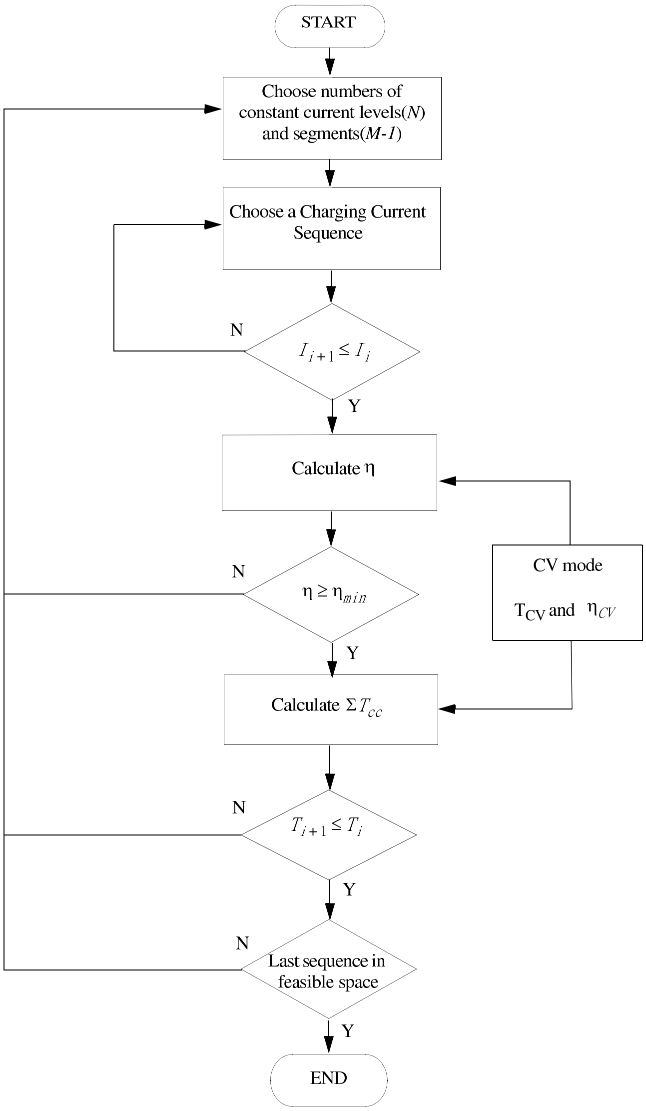

Fig. 4. Multi-step Constant Current Charging Flow Chart

segments are applied in the ILP algorithm to search for optimal charging time and compute the charging efficiency as well. The results are shown in TABLE. VIII. This experiment reveals a fact that a number of SoC segments greater than certain number (e.g. ten in this case) will not help much in improvement of charging time reduction, as shown in Fig. 6. Also, the difference between applying different numbers of charging current levels is almost negligible while the number of charging current levels is more than 5. The result is shown in TABLE. IX.

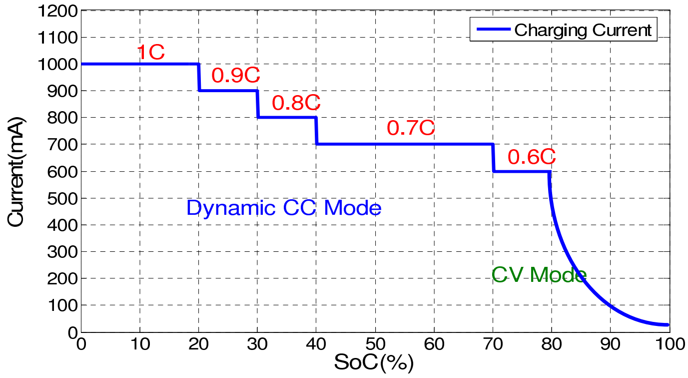

Fig. 5. Optimal Charging Profile with Multiple CV Stages

TABLE VIII
EXPERIMENTAL RESULTS OF DIFFERENT SoC SEGMENTS

Segments	CC-CV	4	5	10	20
Charging Time	179.67	158.97	149.47	146.87	146.84
Improvement	-	11.5%	17.3%	18.25%	18.27%

B. Fast and Slow charging

Both fast and slow charging regimes from a commercial product specification are used to compare with the proposed algorithm. Fast and slow charging are to charge the battery with 1C-rate and 0.3C-rate till 80% SoC, respectively. By applying ILP in searching the optimal charging profile, the experimental results show that the fast charging time is reduced from 179.67 minutes to 146.87 minutes, and the slow charging time is reduced from 275.83 minutes to 216.97 minutes as shown in TABLE X. The improvement rates are 18.25% and 21.38%, respectively.

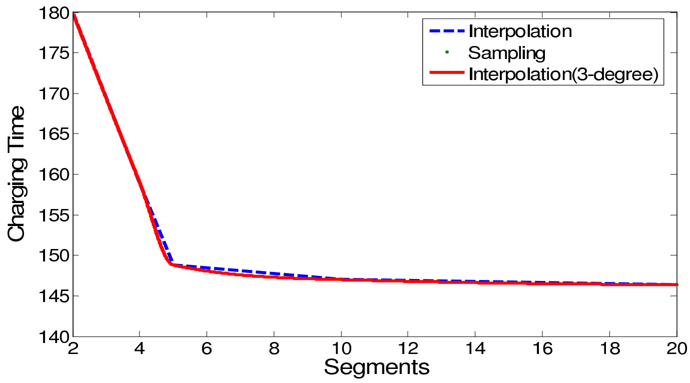

Fig. 6. Improvement of Charging time for different numbers of SoC segments

TABLE IX
COMPARISON RESULTS OF DIFFERENT CHARGING CURRENT LEVELS

Currents	CC-CV	5	10
Charging Time	179.67	148.96	146.87
Improvement	-	17.09%	18.25%

TABLE X
COMPARISON WITH DIFFERENT CHARGING SPEEDS

Charging Speed	Fast Charging	Slow Charing
CC-CV	179.67 mins	275.83 mins
Multi-Level CC-CV with ILP Scheduling	146.87 mins	216.97 mins
Improvement	18.25%	21.38%

V. CONCLUSION

A novel algorithm applying $(0, 1)$-integer linear programming is proposed to search the optimal charging profile for rechargeable Lithium-ion batteries. Multi-step (level) constant current method is applied in the charging experiments. The grade of charging currents and number of segments are also discussed. Here we proposed a novel algorithm using $(0, 1)$-integer linear programming(ILP) to search the optimal charging profile for multi-step constant current and constant voltage charging scheme. Experimental results show that the proposed algorithm achieves up to 18.25% and 21.38% charging time reduction in comparison to fast- and slow-charging specifications of a commercial product respectively, while the charging efficiency is constrained at an acceptable level.

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REFERENCES

[1] R. A. Powers, "Batteries for low power electronics," in *Proc. of the IEEE*, vol. 83, no. 4, April 1995, pp. 687–693.
[2] A. W. Swager, "Smart battery technology: Power management's missing link," *EDN*, pp. 47–64, March 1995.
[3] L. J. Curran, "Charger ics reflect shift to smart batteries," *EDN*, pp. 10–11, 1997.
[4] R. Schweber, "Supervisory ics empower batteries to take charge," *EDN*, pp. 61–72, September 1997.
[5] D. Linden, *Handbook of Batteries*, 3rd ed. McGraw-Hill, 2002.
[6] T. Ikeya, N. Sawada, S. Takagi, J. Murakami, K. Kobayashi, T. Sakabe, E. Kousaka, H. Yoshioka, S. Kato, M. Yamashita, H. Narisoko, Y. Mita, K. Nishiyama, K. Adachi, and K. Ishihara, "Multi-step constant-current charging method for an electric vehicle, valve-regulated, lead-acid batteries during night time for load-leveling," *Journal of Power Sources*, vol. 75, no. 1, pp. 101–107, September 1998.
[7] T. Ikeya, N. Sawada, J. Murakami, K. Kobayashi, M. Hattori, N. Murotani, S. Ujiie, K. Kajiyama, H. Nasu, H. Narisoko, Y. Tomaki, K. Adachi, Y. Mita, and K. Ishihara, "Multi-step constant-current charging method for an electric vehicle nickel/metal hybrid battery with high-energy efficiency and long cycle time," *Journal of Power Sources*, vol. 105, no. 1, pp. 6–12, March 2002.
[8] Y. Liu, J. Teng, and Y. Lin, "Search for an optimal charging pattern for lithium-ion battery using ant colony system algorithm," *IEEE Tran. on Industrial Electronics*, vol. 52, no. 5, pp. 1328–1336, October 2005.
[9] A. Schrijver, *Theory of Linear and Integer Programming*. John Wiley & Sons, 1986.
[10] L. A. Wolsey, *Integer Programming*. John Wiley &Sons, 1998.
[11] H. Bergveld, W. Kruijt, and P. Notten, *Battery Management Systems Design by Modelling*. Kluwer Academic Publishers, 2002, vol. 1.
[12] V. Pop, H. Bergveld, P. Danilov, and R. Regtien, *Battery Management Systems-Accurate State of Charge Indication for Battery-Power Applications*. Springer, 2008.